

DevOps Series

Using Ansible to Set Up HAProxy as a Load Balancer for Nginx

In this 24th article in the DevOps series, we will learn how to set up HAProxy as a load balancer for multiple Nginx Web servers using Ansible.



HAProxy is free, open source, highly available, load balancer software written by Willy Tarreau in 2000. It is implemented in the C programming language. It is known for its high performance and is extremely reliable and secure. It supports both Layer 4 (TCP) and Layer 7 (HTTP) based application load balancing, and is released under the GPLv2 licence. Nginx is a Web server created by Igor Sysoev, and is also written in the C programming language. It can be used as a reverse proxy, mail proxy and as an HTTP cache. It was first released in 2004 and uses the 2-clause BSD licence.

Setup

The HAProxy and Nginx installations use CentOS 7 (x86_64) as their base operating system. A single instance is launched using KVM for running HAProxy. A couple of CentOS VMs are provisioned to install and configure Nginx. The *centos* users in all the VMs are given sudo access using the *visudo* command. SELinux is disabled for the exercise.

The host system is a Parabola GNU/Linux-libre x86_64 system and Ansible is installed using the distribution package manager. The version of Ansible used is 2.6.0 as indicated below:

```
$ ansible --version
ansible 2.6.0
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/guest/.ansible/
plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.6/site-
packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.6.5 (default, May 11 2018, 04:00:52)
[GCC 8.1.0]
```

The Ansible inventory, files and playbook directories are created on the host system as follows:

```
ansible/inventory/kvm/
  /playbooks/configuration/
  /files/
```

The *inventory/kvm/inventory* file contains the following:

```
[front]
haproxy ansible_host=192.168.122.113 ansible_connection=ssh
ansible_user=centos ansible_password=password

[web1]
nginx1 ansible_host=192.168.122.248 ansible_connection=ssh
ansible_user=centos ansible_password=password

[web2]
nginx2 ansible_host=192.168.122.147 ansible_connection=ssh
ansible_user=centos ansible_password=password

[web:children]
web1
web2
```

The 'front' group contains the HAProxy instance information. The couple of Nginx Web servers are grouped together under a 'web' group, and can also be accessed individually. You can test connectivity from Ansible to the CentOS guest VMs using the following Ansible commands:

```
$ ansible -i inventory/kvm/inventory haproxy -m ping
haproxy | SUCCESS => {
  "changed": false,
  "ping": "pong"
}

$ ansible -i inventory/kvm/inventory nginx1 -m ping
nginx1 | SUCCESS => {
```